Geometric Structural Limits

A State-Space Formulation of Non-Derivable Structural Conditions

Abstract

Many dynamical systems evolve within domains whose structural limits are not determined by the internal dynamics itself. This paper introduces a geometric framework for analyzing such limits in state space. We define an admissible structural domain $\Omega_S \subset \Omega$, its boundary ∂S, and a structural distance function measuring proximity to collapse. We prove that the structural boundary cannot, in general, be derived solely from the internal vector field governing system evolution. This result is interpreted within the framework of the **Theory of Axiomatic Necessity (TNA)**, which posits that operational systems require structural conditions that cannot be internally derived. The geometric formulation presented here provides a quantitative representation of proximity to structural rupture across physical, biological, and computational systems.

1 Introduction

Many natural and artificial systems operate within structural domains that constrain their admissible states. While the internal dynamics governs evolution within such domains, the limits of admissibility themselves often arise from external or higher-order structural conditions.

Examples include:

- phase boundaries in physical systems
- extinction thresholds in ecological dynamics
- training collapse in machine learning systems.

These phenomena suggest a distinction between **internal dynamical laws** and **structural admissibility conditions**.

In formal logic, analogous limitations were demonstrated by Kurt Gödel, whose incompleteness results show that sufficiently expressive formal systems contain truths that cannot be derived from their internal axioms alone. Inspired by this idea, the **Theory of Axiomatic Necessity (TNA)** proposes that operational systems require at least one structural condition that cannot be derived from internal dynamics.

This paper provides a **geometric formulation** of such structural limits in dynamical systems.

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2 Structural Admissibility Domains

Let

$$\Omega \subseteq \mathbb{R}^n$$

be a state space.

Define a structural function

$$\phi : \Omega \to \mathbb{R}$$

and the **admissible structural domain**

$$\Omega_S = \{\omega \in \Omega \mid \phi(\omega) \geq 0\}.$$

The **structural boundary** is

$$\partial S = \{\omega \in \Omega \mid \phi(\omega) = 0\}.$$

The domain Ω_S represents the set of states for which the system remains operational.

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3 Dynamical System

Consider a dynamical system defined by

$$\dot{\omega} = F(\omega)$$

where

$$F : \Omega_S \to \mathbb{R}^n$$

is a vector field defined on the admissible domain.

We assume that

- F is **continuously differentiable** on Ω_S
- F is **locally Lipschitz**.

No smooth extension to the structural boundary ∂S is required. In many systems the vector field may exhibit discontinuities or singular behavior near the boundary, representing regime transitions or structural collapse.

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4 Structural Distance to Collapse

To quantify proximity to structural rupture, define the **structural distance function**

$$d_S(\omega) = \inf_{\omega' \in \partial S} d(\omega, \omega').$$

The metric d may be chosen according to the geometry of the state space. In Euclidean systems the standard Euclidean distance is appropriate, while in statistical manifolds alternative metrics such as the Fisher information metric may be used.

As trajectories approach the boundary,

$$d_S(\omega(t)) \to 0,$$

indicating increasing proximity to structural collapse.

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5 Derivability from Internal Dynamics

Let F denote the internal vector field governing the system.

Define

$$D(F)$$

as the set of subsets of Ω that can be characterized using functionals constructed solely from F and its derivatives.

Formally,

$$D(F) = \{A \subseteq \Omega \mid A = \{\omega : G(F, \nabla F, \ldots) \geq 0\}\}$$

for some functional G.

This set represents structural properties that are **derivable from internal dynamics**.

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6 Theorem (Structural Non-Derivability of Domain Boundaries)

Let

$$F : \Omega_S \to \mathbb{R}^n$$

be a continuously differentiable vector field defined on an admissible domain

$$\Omega_S \subset \Omega \subset \mathbb{R}^n$$

where the structural boundary is defined by a function

$$\phi : \Omega \to \mathbb{R}$$

such that

$$\Omega_S = \{\omega \in \Omega \mid \phi(\omega) \geq 0\}$$

and

$$\partial S = \{\omega \in \Omega \mid \phi(\omega) = 0\}.$$

Assume that the vector field F contains no explicit dependence on the structural function ϕ.

Define the set of dynamically derivable subsets

$$D(F) = \{A \subseteq \Omega \mid A = \{\omega : G(F, \nabla F, \ldots) \geq 0\}\}.$$

Then, in general,

$$\partial S \notin D(F).$$

Demonstration

The vector field F is defined only on the admissible domain Ω_S.
All functionals $G(F, \nabla F, \ldots)$ therefore depend exclusively on information contained within the internal dynamics of this domain.

Consequently, any set $A \in D(F)$ must be determined solely by properties of the vector field and its derivatives evaluated on states belonging to Ω_S.

However, the structural boundary ∂S is defined by the zero level set of the structural function ϕ , which determines the admissibility of states themselves rather than their internal evolution.

Since ϕ is independent of the vector field F, knowledge of F and its derivatives within Ω_S does not uniquely determine the level set $\phi = 0$.

Indeed, infinitely many distinct structural functions ϕ_i can produce the same internal vector field F on Ω_S while generating different boundaries ∂S_i.

Therefore the structural boundary cannot, in general, be reconstructed from functionals of F alone.

Hence

$$\partial S \notin D(F).$$

7 Geometric Interpretation

The theorem admits a simple geometric interpretation.

A trajectory

$$\omega(t)$$

evolves within the admissible domain Ω_S. As the trajectory approaches the boundary,

$$d_S(\omega) \to 0.$$

At the collapse point

$$\omega_c \in \partial S$$

the dynamical flow becomes non-extensible, indicating structural rupture.

The geometric situation is illustrated by a trajectory approaching the structural boundary in state space.

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8 Relation to the Theory of Axiomatic Necessity (TNA)

The result aligns with the **Theory of Axiomatic Necessity**, which distinguishes between two structural levels:

COMPONENT	ROLE
Internal dynamics	governs evolution within the domain
Structural conditions	define the domain itself

In earlier work, structural admissibility was described through a binary operator

$$S : \Omega \to \{0, 1\}$$

where $S(\omega) = 1$ indicates admissibility.

The domain introduced here corresponds to

$$\Omega_S = S^{-1}(1).$$

The structural boundary

$$S(\omega) = 0$$

represents transition to the non-operational regime N_1.

The geometric distance $d_S(\omega)$ introduced in this paper provides a **continuous measure of proximity to this structural transition**.

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9 Implications for Complex Systems

The geometric framework developed here applies broadly to systems exhibiting regime transitions.

Examples include:

- phase transitions in physics
- ecological collapse thresholds
- training instability in neural networks
- financial systemic failures.

In each case, the internal dynamics governs evolution within the operational domain, while structural conditions determine the limits of that domain.

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10 Conclusion

We introduced a geometric framework for analyzing structural limits in dynamical systems. By defining structural admissibility domains and a distance-to-collapse metric, we demonstrated that structural boundaries cannot generally be derived from internal dynamics alone.

This result supports the broader principle of **axiomatic necessity**, according to which operational systems require structural conditions that lie beyond the scope of their internal dynamical laws.

Graphic interpretation

$$dS(\omega) = \omega\prime \in \partial Sinf \parallel \omega - \omega\prime \parallel$$

$$\omega(t) \to \omega_c lim_{t \to t_c} \omega(t) = \omega_c$$

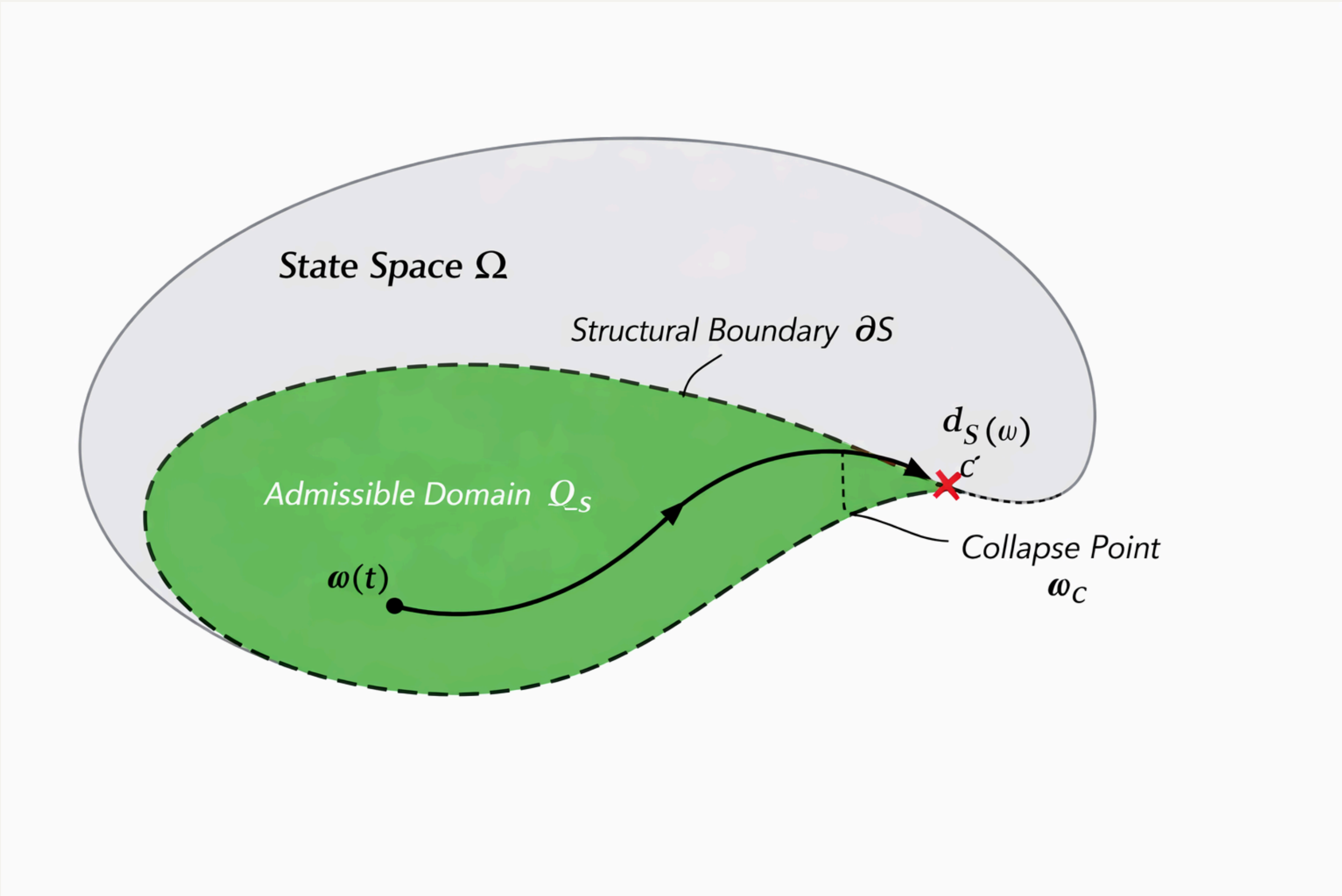

Figure 1 — Structural Collapse in State Space.
A trajectory ω(t) evolves inside the admissible domain Ω*S of the state space Ω.*
The dashed curve represents the structural boundary ∂S.
As the trajectory approaches the boundary, the structural distance d_S(ω) tends to zero.
*At the collapse point ω*c the dynamical flow becomes non-extensible, indicating a structural rupture of the system.

The geometric interpretation of the theorem is illustrated in Figure 1.
A trajectory ω(t) evolves inside the admissible domain Ω*S until it approaches the structural boundary ∂S.*
When the structural distance d_S(ω) tends to zero, the dynamical regime reaches a point where the

governing vector field cannot be extended.
The trajectory therefore terminates at a collapse point ω_c.